

iKwath

Proposed Technical Solution

A formulation-aware, pod-based, feedback-controlled appliance for fresh standardized Kwatha preparation

DESIGN TARGET

Fresh single-dose Kwatha on demand;
approximately 400 mL process water to
approximately 100 mL final dose.

CORE ENGINEERING IDEA

Minimize cycle time by improving heat/mass
transfer and endpoint control — not by simply
increasing temperature.

QUALITY CONSTRAINT

Preserve the relevant qualitative constituent
profile and quantitative extractive yield of the
validated reference preparation.

CONTROL STRATEGY

Pod recipe + temperature + weight + pressure +
density/solids endpoint + safety interlocks +
batch logging.

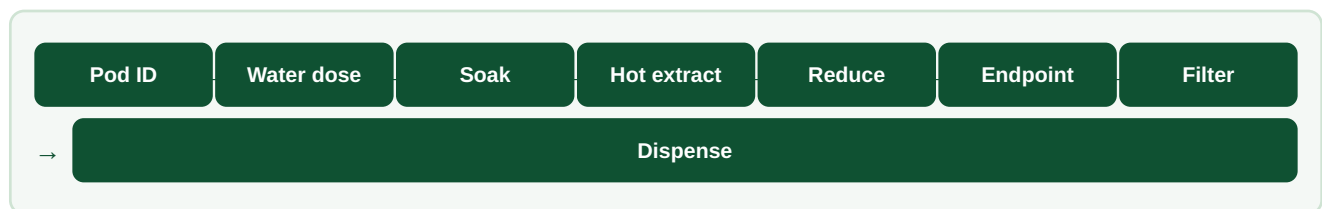
1. Executive Summary

The proposed iKwath system is a compact countertop appliance that converts a single-dose pod of standardized coarse Ayurvedic powder (yavakuṭa cūrṇa) into a fresh Kwatha with minimal user intervention. The machine identifies the formulation, meters the required water, performs controlled hot extraction, reduces the liquid to the prescribed endpoint, filters the decoction and dispenses one fresh dose.

The central innovation is a mild-temperature, closed-loop acceleration strategy. Instead of shortening the cycle through excessive heating, iKwath improves mass transfer and evaporation using formulation-specific agitation/recirculation, a high-area evaporation geometry, controlled vapour removal and, where validated, reduced pressure. Weight and extract-density/solids measurements are used to detect the endpoint instead of relying only on a fixed timer.

SIH SUCCESS CRITERION

The SIH success criterion is not simply "fast Kadha." It is: the fastest practically achievable process that still reproduces the validated reference decoction's relevant constituent profile and extractive yield.



Why this approach is strong

- **Formulation-aware:** one generic recipe is not assumed for all Kwathas.
- **Closed-loop:** the machine measures actual process state and responds to it.
- **Mass-transfer focused:** the time bottleneck is attacked without relying on high temperature.
- **Validation-first:** acceleration settings are accepted only after comparison with a reference preparation.
- **Hygiene by design:** disposable spent-powder pod plus removable wet-contact parts and rinse cycle.

2. Problem Definition and Design Requirements

The SIH statement is best understood as a controlled extraction and reduction problem, not as a generic cooking appliance. The user should only need to add water and insert a formulation pod; the machine must execute the preparation consistently.

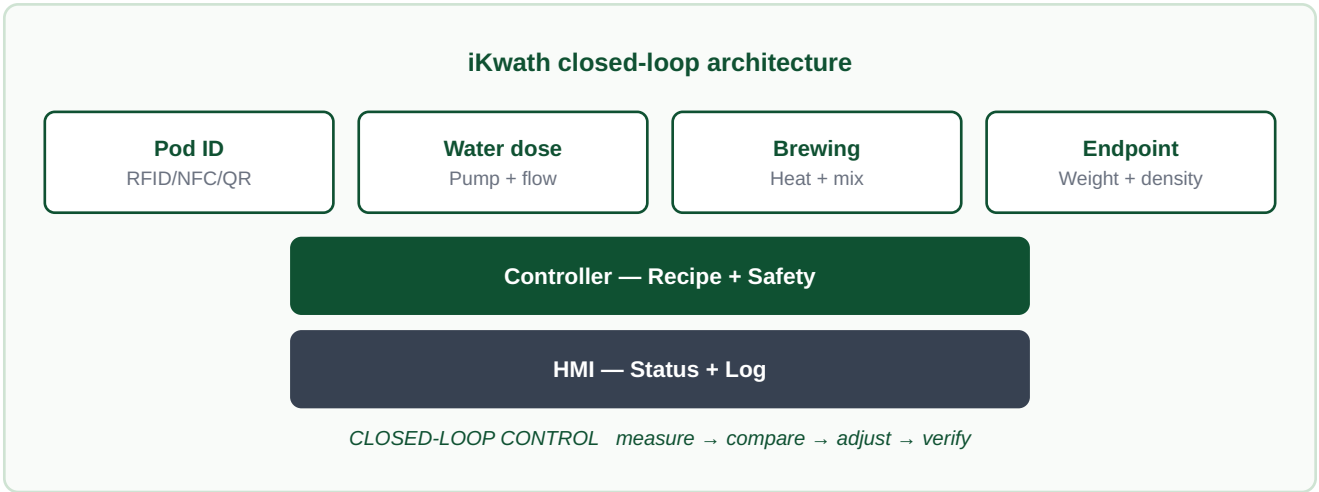
SIH requirement	Meaning	Engineering response
Fresh on demand	Prepare immediately before consumption.	Single-dose batch; direct dispense.
Pod-based	Standardized coarse powder is pre-portioned and contained.	Brew-bag pod + formulation ID.
AFI/API profile	Formulation determines water, process and endpoint requirements.	Locked digital recipe linked to pod ID.
~400 mL → ~100 mL	Large reduction is required before dispensing.	Mass-based reduction control.
~85–90 °C	Acceleration cannot simply come from high-temperature cooking.	Closed-loop mild-temperature control; vacuum for evaporation if validated.
Quality + yield	Fast output must remain comparable to reference.	Analytical fingerprint + extractive-yield validation.
Easy cleaning	Residues must not create excessive manual work.	Removable chamber/filter + rinse cycle.

CRITICAL ENGINEERING OBSERVATION

At ordinary atmospheric pressure, water boils near 100 °C at sea level. Therefore, a true boiling/reduction stage around 85–90 °C requires sub-atmospheric pressure. This makes controlled vacuum-assisted evaporation a technically coherent option for the SIH specification. The prototype should directly measure absolute pressure and experimentally validate the pressure–temperature operating window.

The proposed process separates hot extraction from mild-temperature reduction. This gives the team a controllable way to accelerate water removal without resorting to aggressive heating.

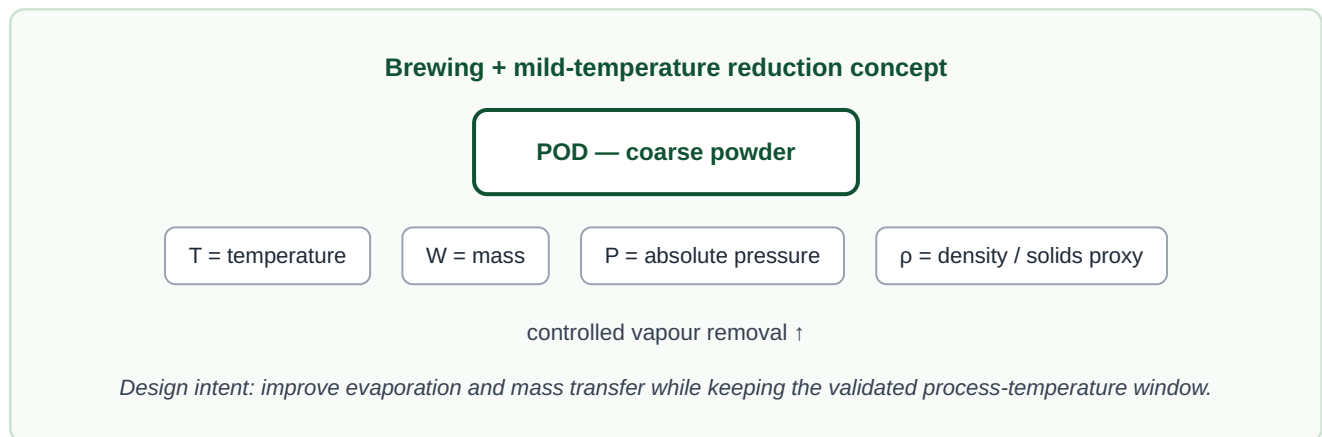
3. Proposed System Architecture



Core modules

Module	Function	Prototype direction
Smart pod interface	Identify formulation and keep coarse powder contained.	RFID/NFC; printed code as fallback; removable pod basket.
Water dosing	Deliver formulation-defined water amount.	Food-grade reservoir + pump + flow meter; gravimetric check.
Extraction chamber	Controlled wetting, temperature and contact.	316L stainless steel; removable pod holder; gentle agitation.
Reduction module	Remove water efficiently at validated mild temperature.	Shallow/high-area surface + vapour removal + optional regulated vacuum.
Endpoint sensing	Detect actual process completion.	Load cell + pressure + temperature + density/solids measurement.
Controller	Run recipe, feedback loops and safety logic.	ESP32/STM32-class prototype; PLC for productization.
HMI/logging	Show stage, endpoint and faults.	Display + optional mobile dashboard; batch record.

4. End-to-End Working Principle



1. **Pod verification.** Read formulation ID and load its controlled preparation profile: water amount, temperature window, extraction sequence, reduction target and tolerances.
2. **Water dosing.** Pump the required water quantity and verify delivery using flow plus chamber weight.
3. **Soak / hydrate.** Pre-wet the coarse powder. Gentle agitation improves wetting and reduces stagnant zones.
4. **Controlled extraction.** Maintain the formulation-specific temperature profile using closed-loop heater control; use low-shear agitation/recirculation to improve contact and uniformity.
5. **Accelerated reduction.** Expose the liquid to a larger effective evaporation area and remove vapour gently. If validated, regulate vacuum so evaporation occurs within the mild temperature window.
6. **Endpoint verification.** Track chamber mass and a calibrated density/solids signal. Do not terminate solely on elapsed time.
7. **Final filtration.** Retain coarse spent material in the pod and pass the liquid through a final food-grade filter.
8. **Dispense + rinse.** Deliver the fresh dose, record the batch, remove the pod and run the rinse sequence.

5. Technical Strategy for Minimizing Preparation Time

The amount of water to be removed is fixed by the formulation, so the engineering opportunity is to increase the rate of heat and mass transfer while preserving the required process temperature, water proportion and reduction endpoint.

Acceleration lever	Mechanism	Constraint / validation
High-area evaporation surface	More liquid surface is available for evaporation at the same temperature.	Verify temperature uniformity and output chemistry.
Gentle vapour removal	Maintains an evaporation driving force by removing vapour from above the liquid.	Control flow to avoid foaming/entrainment; validate volatile-loss risk.
Controlled vacuum	Lowers boiling temperature and can intensify evaporation at ~85–90 °C.	Pressure stability and formulation-specific validation required.
Gentle agitation / recirculation	Improves wetting, temperature uniformity and solid–liquid mass transfer.	Optimize speed; excessive shear may increase fines/foaming.
Feedback endpoint	Stops at the actual reduction/concentration endpoint rather than a fixed time.	Sensor must be calibrated against laboratory measurements.

Recommended acceleration stack

Extraction: controlled hot extraction + gentle agitation/recirculation. Reduction: shallow/high-area evaporation + controlled vapour removal, with regulated vacuum when validated. Endpoint: weight + density/solids + temperature history. This combination attacks both extraction efficiency and the water-removal bottleneck without changing the formulation's prescribed water proportion.

Do not make an unverified claim

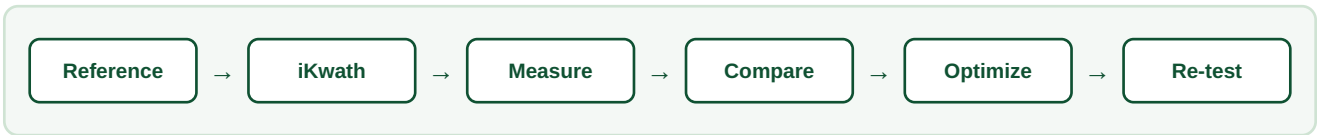
Do not make an unverified "5-minute" claim. Use an engineering target such as ≤ 20 minutes for a selected benchmark formulation only as a prototype optimization target; the final SIH claim should use measured cycle time together with analytical equivalence and repeatability.

6. Qualitative and Quantitative Quality Preservation

A successful accelerated process must match the reference preparation in two dimensions: qualitative chemical profile and quantitative extractive yield.

Dimension	Measurement concept	Acceptance approach
Constituent profile	Chromatographic/spectroscopic fingerprint and relevant marker compounds, as appropriate.	Predefined analytical similarity criteria established with experts/lab.
Extractive yield	Validated laboratory measurement of extractive matter/solids.	Within formulation-specific reference range.
Final dose	Delivered volume/mass.	Consistent single-dose delivery.
Temperature history	Continuous temperature curve.	Inside validated formulation-specific window.
Reduction endpoint	Mass loss + density/solids endpoint.	Target reached without over-reduction.
Repeatability	Replicate runs with controlled pod lots.	Low batch-to-batch variation.

Reference-vs-iKwath optimization loop



- **Reference:** prepare replicate classical/reference batches under controlled conditions.
- **Design of experiments:** vary agitation, surface area, vapour removal and vacuum profile within safe limits.
- **Analytical comparison:** compare chemical fingerprint/markers and extractive yield.
- **Lock the recipe:** select the fastest configuration that remains inside the predefined acceptance window.
- **Repeat:** verify across multiple runs and later across pod lots.

This validation loop directly answers the SIH requirement that acceleration must not alter the final decoction.

7. Control System, Sensors and Hardware

Component	Role	Prototype direction
RTD (PT100/PT1000)	Accurate process-temperature feedback.	Food-compatible probe.
Load cell	Water-dose verification and reduction tracking.	Platform beneath removable chamber.
Absolute pressure sensor	Vacuum control and safety monitoring.	Rated for hot humid vapour service.
Density / solids sensor	Endpoint/concentration signal.	Inline density or refractive-index sensing after suitable filtration; formulation-calibrated.
Flow meter	Water and optional recirculation measurement.	Food-compatible flow sensor.
Heater	Controlled thermal input.	Resistive heater or suitable induction vessel.
Agitator	Mass-transfer enhancement.	Low-shear magnetic or sealed mechanical drive.
Vacuum pump + condenser	Mild-temperature evaporation and vapour capture.	Small food-process-compatible pump, condenser and trap.
Safety interlocks	Dry-run, overtemperature, overpressure, lid-open and boil-over protection.	Independent hardware cutoff + software fault states.

Control logic

Use a state machine, with the safety layer given priority. The recipe layer supplies formulation-specific setpoints; the process layer controls heater, pump, agitator and vacuum; the sensing layer supplies actual state.

IDLE → IDENTIFY → DOSE → SOAK → EXTRACT → REDUCE → VERIFY → FILTER/DISPENSE → RINSE

The mobile app, if used, should remain a monitoring/traceability layer rather than the safety controller. The appliance must be able to safely complete or abort a cycle without a phone connection.

8. Pod Design, Hygiene and Safety

The pod is the controlled interface between standardized raw material and the appliance. A single-dose pod reduces manual weighing and keeps spent coarse material contained.

Pod element	Proposed design	Purpose
Powder chamber	Pre-weighed standardized coarse powder.	Controls dose and powder grade.
Brew-bag membrane	Food-grade porous material selected for wet strength and suitable flow.	Allows water contact while retaining solids.
Identifier	RFID/NFC plus printed secondary code.	Maps pod to formulation and lot.
Handling tab	Dry external tab.	Tool-free insertion/removal.
Lot metadata	Formulation, lot, expiry, recipe version.	Traceability and controlled recipe updates.

Cleaning architecture

- Removable dishwasher-safe or otherwise validated-cleanable boiling chamber and final filter.
- Smooth food-grade stainless-steel wet-contact surfaces with minimal crevices.
- Disposable spent-powder pod to reduce manual residue handling.
- Automated rinse cycle after dispensing; formulation-changeover rules.

Safety architecture

- Lid/door interlock before heating or vacuum activation.
- Independent thermal cutoff and software overtemperature alarm.
- Pressure sensor + mechanical relief path; controlled vacuum ramp.
- Dry-run and low-liquid protection.
- Boil-over/foam mitigation through headspace and conservative process control.

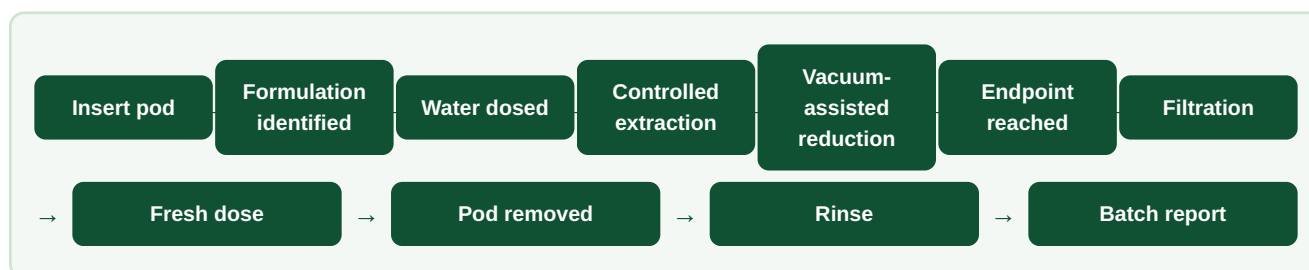
9. Validation, Innovation and SIH Demonstration

Test	Metric	Demonstration method
Cycle time	Total and stage-wise time.	Reference vs baseline vs accelerated prototype.
Temperature	Actual curve and stability.	Log RTD data throughout extraction/reduction.
Reduction	Mass vs time and endpoint repeatability.	Show load-cell trace.
Density endpoint	Sensor value vs lab value.	Build formulation-specific calibration.
Chemical equivalence	Fingerprint / markers.	External laboratory comparison of replicate samples.
Extractive yield	Validated analytical result.	Reference vs iKwath batches.
Cleaning	Residue and cleaning time.	Removable parts + rinse demonstration.
Safety	Fault response.	Test dry run, lid-open, overtemperature and pressure faults.

Innovation / USP

"iKwath is a formulation-aware, closed-loop decoction appliance that accelerates the rate-limiting extraction/reduction stages at a validated mild process temperature, while using measurable endpoints to reproduce the reference decoction rather than merely reproducing a cooking time."

SIH live demo sequence



Development roadmap

Phase	Deliverable
1. Reference study	Select 1–2 benchmark formulations; establish reference time, yield and analytical profile.
2. Instrumented chamber	Build 1 L chamber with RTD, load cell, pressure sensor and agitation.
3. Acceleration trials	Compare surface area, vapour removal, agitation and vacuum profiles.
4. Closed-loop prototype	Implement recipe state machine, endpoint control, pod ID and safety interlocks.
5. Validation	Replicate reference vs iKwath batches and lock the fastest acceptable recipe.
6. Productization	Improve enclosure, manufacturability, cleaning, pod supply chain and compliance pathway.

Technical basis / source notes

- PCIM&H — official portal — Ayurvedic Pharmacopoeia monographs and formulations: <https://www.portal.pcimh.gov.in/products>
- PCIM&H — Formulary Ayurvedic Kwatha Curna, General Notices: <https://pcimh.gov.in/WriteReadData/RTF1984/FormularyAyushKvathaCurna.pdf>
- PCIM&H — API Jala monograph: https://www.portal.pcimh.gov.in/product_details/993203a5-502a-4afa-88dd-75acdb437531
- Ministry of Ayush — API/AFI and PCIM&H standards: <https://ayush.gov.in/resources/pdf/Parliament/winter2022R.pdf>
- Muala et al. (2021), aqueous decoction optimization: <https://pubmed.ncbi.nlm.nih.gov/33912716/>
- Pelargonium graveolens aqueous decoction optimization: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6260416/>
- Assessment of decoction methods and extractive yield: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3609273/>
- Review — Advances in Natural Product Extraction, decoction section: <https://pmc.ncbi.nlm.nih.gov/articles/PMC13074414/>

Important note

Vacuum, stirring, density/refractive-index sensing and other hardware choices are engineering hypotheses to be validated. They are not automatically equivalent to classical preparation. The acceptance criterion is formulation-specific output equivalence under the applicable official specification and expert/laboratory validation.

Final proposal in one sentence:

iKwath will make fresh single-dose Kwatha by combining standardized pods, formulation-aware recipes, controlled mild-temperature extraction, high-area/vacuum-assisted reduction and closed-loop endpoint sensing — then prove the fastest acceptable process by analytical comparison with the classical reference.